1.1. Formal Definition of Machine Learning

Machine Learning (ML) is neither an all-purpose optimization engine nor an infallible heuristic. Formally, Machine Learning is defined as:

$$\textbf{An engineering approach to } \underbrace{\textbf{learn}}_{(1)}\underbrace{\textbf{complex}}_{(2)}\underbrace{\textbf{patterns}}_{(3)} \textbf{ from } \underbrace{\textbf{existing data}}_{(4)} \textbf{ and use these patterns to make } \underbrace{\textbf{predictions}}_{(5)} \textbf{ on } \underbrace{\textbf{unseen data}}_{(6)}.$$

Mapping function $f : X \to Y$ is too complex. Also lookup table is $O(1)$ complexity.

